

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286014

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Lamarra; Michael Christopher et al.

WIRE BONDING ARRAYS, MAINTENANCE STATIONS FOR WIRE BONDING ARRAYS, AND RELATED METHODS

Abstract

A wire bonding array is provided. The wire bonding array includes a plurality of wire bonding systems, each of the plurality of wire bonding systems including a bond head assembly for bonding a wire to a workpiece and a support structure for supporting the workpiece. The wire bonding array also includes a wire bonding tool replacement system configured for installation on, and removal from, each of the plurality of wire bonding systems. The wire bonding array also includes a service robot configured to install the wire bonding tool replacement system on, and remove the wire bonding tool replacement system from, each of the plurality of wire bonding systems.

Inventors: Lamarra; Michael Christopher (Harleysville, PA), Venna; Manoj Kumar (Singapore, SG)

Applicant: Kulicke and Soffa Industries, Inc. (Fort Washington, PA)

Family ID: 1000008515882

Assignee: Kulicke and Soffa Industries, Inc. (Fort Washington, PA)

Appl. No.: 19/062494

Filed: February 25, 2025

Related U.S. Application Data

us-provisional-application US 63562833 20240308

Publication Classification

Int. Cl.: H01L23/00 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/562,833, filed on Mar. 8, 2024, and U.S. Provisional Application No. 63/650,997, filed on May 23, 2024, the contents of both are incorporated herein by reference.

FIELD

[0002] The invention relates to wire bonding arrays, maintenance stations for wire bonding arrays, and related methods.

BACKGROUND

[0003] In the processing and packaging of semiconductor devices, wire bonding continues to be a primary method of providing electrical interconnection between two locations within a package (e.g., between a die pad of a semiconductor die and a lead of a leadframe). More specifically, using a wire bonder (also known as a wire bonding system) wire loops are formed between respective locations to be electrically interconnected. The primary methods of forming wire loops are ball bonding and wedge bonding. In forming the bonds between (a) the ends of the wire loop and (b) the bond site (e.g., a die pad, a lead, etc.) varying types of bonding energy may be used, including, for example, ultrasonic energy, thermosonic energy, thermocompressive energy, amongst others. Wire bonding systems are also used to form conductive bumps from portions of wire, and vertical wire structures (e.g., bonded at a single bond site).

[0004] Replacement of a wire bonding tool on a wire bonding system (and/or other related maintenance) often results in significant delays or machine downtime, and utilizes an operator to accomplish the replacement/maintenance. Thus, it would be desirable to provide improved systems and methods for addressing such deficiencies.

SUMMARY

[0005] According to an exemplary embodiment of the invention, a wire bonding array is provided. The wire bonding array includes a plurality of wire bonding systems, each of the plurality of wire bonding systems including (i) a bond head assembly for bonding a wire to a workpiece and (ii) a support structure for supporting the workpiece. The wire bonding array also includes a wire bonding tool replacement system configured for installation on, and removal from, each of the plurality of wire bonding systems. The wire bonding array also includes a service robot configured to install the wire bonding tool replacement system on, and remove the wire bonding tool replacement system from, each of the plurality of wire bonding systems.

[0006] According to other embodiments of the invention, the wire bonding array recited in the immediately preceding paragraph may have any one or more of the following features: each of the plurality of wire bonding systems includes a docking station configured to receive the wire bonding tool replacement system; the docking station includes a connector for providing power and data to the wire bonding tool replacement system; the docking station includes locator pins for aligning the wire bonding tool replacement system during installation; at least one additional wire bonding tool replacement system configured for installation on, and removal from, each of the plurality of wire bonding systems; and the service robot includes a base for moving the service robot with respect to the plurality of wire bonding systems, the base being positioned above a receiving portion of each of the plurality of wire bonding systems configured to receive the wire bonding tool replacement system.

[0007] According to another exemplary embodiment of the invention, a wire bonding array is provided. The wire bonding array includes a plurality of wire bonding systems, each of the

plurality of wire bonding systems including (i) a bond head assembly for bonding a wire to a workpiece and (ii) a support structure for supporting the workpiece. The wire bonding array also includes a maintenance station including a plurality of wire bonding tool replacement systems, each of the plurality of wire bonding tool replacement systems being configured for installation on, and removal from, each of the plurality of wire bonding systems. The wire bonding array also includes a service robot configured to move ones of the plurality of wire bonding tool replacement systems between the maintenance station and ones of the plurality of wire bonding systems.

[0008] According to other embodiments of the invention, the wire bonding array recited in the immediately preceding paragraph may have any one or more of the following features: each of the plurality of wire bonding systems includes a docking station configured to receive each of the plurality of wire bonding tool replacement systems; the maintenance station includes a plurality of docking stations, each of the plurality of docking stations configured to receive each of the plurality of wire bonding tool replacement systems; the maintenance station includes a replacement assembly configured to remove a waste from each of the plurality of wire bonding tool replacement systems, the waste including at least one of (i) an empty wire bonding tool tray configured for holding a plurality of wire bonding tools and (ii) at least one used wire bonding tool; the maintenance station further includes a receptacle for receiving the waste from the plurality of wire bonding tool replacement systems from the replacement assembly; the maintenance station further includes a plurality of wire bonding tool trays, each of the plurality of wire bonding tool trays including a plurality of wire bonding tools; the maintenance station includes a replacement assembly configured to load one of the plurality of wire bonding tool trays into one of the plurality of wire bonding tool replacement systems; the maintenance station further includes a charging station for the service robot; the service robot is further configured to move ones of the plurality of wire bonding tool replacement systems directly between ones of the plurality of wire bonding systems; and the service robot includes a base for moving the service robot with respect to the plurality of wire bonding systems, the base being positioned above a receiving portion of each of the plurality of wire bonding systems configured to receive each of the plurality of wire bonding tool replacement systems.

[0009] According to yet another exemplary embodiment of the invention, a maintenance station for servicing a wire bonding array is provided. The maintenance station includes a plurality of wire bonding tool replacement systems, each of the plurality of wire bonding tool replacement systems being configured for installation on, and removal from, a wire bonding system.

[0010] According to other embodiments of the invention, the maintenance station recited in the immediately preceding paragraph may have any one or more of the following features: a plurality of docking stations configured to receive any one of the plurality of wire bonding tool replacement systems; a replacement assembly configured to remove a waste from each of the plurality of wire bonding tool replacement systems, the waste including at least one of (i) an empty wire bonding tool tray configured for holding a plurality of wire bonding tools and (ii) at least one used wire bonding tool; a receptacle for receiving the waste from the plurality of wire bonding tool replacement systems; a plurality of wire bonding tool trays, each of the plurality of wire bonding tool trays including a plurality of wire bonding tools; a replacement assembly configured to load one of the plurality of wire bonding tool trays into one of the plurality of wire bonding tool replacement systems; a service robot configured to move ones of the plurality of wire bonding tool replacement systems between the maintenance station and ones of a plurality of wire bonding systems; a charging station for the service robot; and the service robot includes a base for moving with respect to the plurality of wire bonding systems, the base being positioned above a receiving portion of each of the plurality of wire bonding systems configured to receive each of the plurality of wire bonding tool replacement systems.

[0011] According to yet another exemplary embodiment of the invention, a method of servicing a wire bonding array is provided. The method includes steps of: (a) providing the wire bonding array

that includes (i) a plurality of wire bonding systems, each of the plurality of wire bonding systems including a bond head assembly for bonding a wire to a workpiece and a support structure for supporting the workpiece, the bond head assembly including a wire bonding tool, (ii) a wire bonding tool replacement system configured for installation on, and removal from, each of the plurality of wire bonding systems, and (iii) a service robot configured to install the wire bonding tool replacement system on, and remove the wire bonding tool replacement system from, each of the plurality of wire bonding systems; (b) carrying the wire bonding tool replacement system using the service robot to one of the plurality of wire bonding systems; and (c) installing the wire bonding tool replacement system on the one of the plurality of wire bonding systems using the service robot.

[0012] According to other embodiments of the invention, the method recited in the immediately preceding paragraph may have any one or more of the following features: a step of (d) removing the wire bonding tool replacement system from the one of the plurality of wire bonding systems with the service robot after step (c); a step of (e1) carrying the wire bonding tool replacement system to another of the plurality of wire bonding systems with the service robot after step (d), and a step of (f1) installing the wire bonding tool replacement system on the another of the plurality of wire bonding systems after step (e1); a step of (e2) carrying the wire bonding tool replacement system to a maintenance station with the service robot after step (d), and a step of (f2) installing the wire bonding tool replacement system on a docking station of the maintenance station after step (e2); and step (a) includes providing the service robot including a base for moving the service robot with respect to the plurality of wire bonding systems, the base being positioned above a receiving portion of each of the plurality of wire bonding systems configured to receive the wire bonding tool replacement system.

[0013] According to yet another exemplary embodiment of the invention, a method of servicing a wire bonding array is provided. The method includes steps of: (a) providing the wire bonding array that includes (i) a plurality of wire bonding systems, each of the plurality of wire bonding systems including a bond head assembly for bonding a wire to a workpiece and a support structure for supporting the workpiece, the bond head assembly including a wire bonding tool, (ii) a maintenance station including a plurality of wire bonding tool replacement systems, each of the plurality of wire bonding tool replacement systems being configured for installation on, and removal from, each of the plurality of wire bonding systems; and (iii) a service robot configured to install ones of the plurality of wire bonding tool replacement systems on, and remove from, each of the plurality of wire bonding systems; (b) carrying one of the plurality of wire bonding tool replacement systems from the maintenance station to one of the plurality of wire bonding systems with the service robot; and (c) installing the one of the plurality of wire bonding tool replacement systems on the one of the plurality of wire bonding systems using the service robot.

[0014] According to other embodiments of the invention, the method recited in the immediately preceding paragraph may have any one or more of the following features: a step of (d) removing the one of the plurality of wire bonding tool replacement systems from the one of the plurality of wire bonding systems with the service robot after step (c); a step of (e1) carrying the one of the plurality of wire bonding tool replacement systems to another of the plurality of wire bonding systems with the service robot after step (d), and a step of (f1) installing the one of the plurality of wire bonding tool replacement systems on the another of the plurality of wire bonding systems after step (e1); a step of (e2) carrying the one of the plurality of wire bonding tool replacement systems to the maintenance station with the service robot after step (d), and a step of (f2) installing the one of the plurality of wire bonding tool replacement systems on a docking station of the maintenance station after step (e2); and step (a) includes providing the service robot including a base for moving the service robot with respect to the plurality of wire bonding systems, the base being positioned above a receiving portion of each of the plurality of wire bonding systems configured to receive each of the plurality of wire bonding tool replacement systems.

[0015] According to yet another exemplary embodiment of the invention, a method of operating a maintenance station of a wire bonding array is provided. The method includes steps of: (a) providing the maintenance station including (i) a plurality of wire bonding tool replacement systems, each of the plurality of wire bonding tool replacement systems being configured for installation on, and removal from, a wire bonding system, (ii) a plurality of wire bonding tool trays, each of the plurality of wire bonding tool trays including a plurality of wire bonding tools, and (iii) a replacement assembly configured to load one of the plurality of wire bonding tool trays into one of the plurality of wire bonding tool replacement systems; and (b) installing one of the plurality of wire bonding tool trays in one of the plurality of wire bonding tool replacement systems using the replacement assembly.

[0016] According to other embodiments of the invention, the method recited in the immediately preceding paragraph may have any one or more of the following features: a step of removing an empty wire bonding tool tray from the one of the plurality of wire bonding tool replacement systems with the replacement assembly prior to step (b); and a step of removing at least one used wire bonding tool from a used wire bonding tool receptacle of the one of the plurality of wire bonding tool replacement systems with the replacement assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The invention is best understood from the following detailed description when read in connection with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity. Included in the drawings are the following figures:

[0018] FIGS. 1A-1C are block diagram side views illustrating a wire bonding system in accordance with an exemplary embodiment of the invention;

[0019] FIG. 2A is a block diagram top view illustrating a docking station for a wire bonding tool replacement system in accordance with an exemplary embodiment of the invention;

[0020] FIG. 2B is a block diagram bottom view illustrating a wire bonding tool replacement system corresponding to the docking station in FIG. 2A in accordance with an exemplary embodiment of the invention;

[0021] FIGS. 3A-3B are block diagram top views illustrating a wire bonding tool replacement system useful in illustrating various exemplary embodiments of the invention;

[0022] FIGS. 4A-4E are block diagrams of a wire bonding array illustrating a method of servicing the wire bonding array in accordance with an exemplary embodiment of the invention;

[0023] FIGS. 5A-5E are block diagrams of the wire bonding array of FIGS. 4A-4E illustrating another method of servicing the wire bonding array in accordance with another exemplary embodiment of the invention;

[0024] FIGS. 6A-6D are block diagrams of the wire bonding array of FIGS. 4A-4E illustrating another method of servicing the wire bonding array in accordance with another exemplary embodiment of the invention;

[0025] FIGS. 7A-7E are block diagrams of the wire bonding array of FIGS. 4A-4E illustrating another method of servicing the wire bonding array in accordance with another exemplary embodiment of the invention;

[0026] FIGS. 8A-8C are block diagrams of various wire bonding arrays in accordance with exemplary embodiments of the invention;

[0027] FIGS. 9A-9H are block diagrams of a maintenance station of a wire bonding array illustrating a method of operating the maintenance station in accordance with various exemplary

embodiments of the invention;

[0028] FIG. **10** is a flow diagram illustrating a method of servicing a wire bonding array in accordance with an exemplary embodiment of the invention; and

[0029] FIG. **11** is a flow diagram illustrating a method of operating a maintenance station of a wire bonding array in accordance with another exemplary embodiment of the invention.

DETAILED DESCRIPTION

[0030] As used herein, the term “wire bonding array” refers to a group of wire bonding systems (e.g., within a production environment, such as a factory). The term “wire bonding array” may include wire bonding systems of different types (e.g., wedge bonders and ball bonders), different model numbers, different manufacturers, different processes, etc. Further, a wire bonding array may include wire bonding systems having different functions within the array (e.g., wire bonding different types of wire, different types of workpieces, etc.). Further still, the term “wire bonding array” may include additional machines, systems, or arrays that support the wire bonding array (e.g., computers for monitoring and/or controlling a factory environment, maintenance stations, service robots, etc.).

[0031] As used herein, the term “wire bonding tool tray” is intended to refer to any type of structure that holds a plurality of wire bonding tools.

[0032] Referring now to FIGS. **1A-1C**, an exemplary wire bonding system **100** is illustrated in a simplified form. Wire bonding system **100** includes a bond head assembly **102**, a support structure **104**, and a docking station **112**. Bond head assembly **102** includes a wire clamp **110**, a transducer **108**, and a wire bonding tool **106** held by transducer **108**. A wire **114** is threaded through wire bonding tool **106**. Wire bonding tool **106** is configured to bond a portion of wire **114** to a workpiece **120**. Workpiece **120** may be any type of workpiece configured for wire bonding. An exemplary workpiece includes a semiconductor element (e.g., a semiconductor die, an LED, etc.) disposed on a substrate (e.g., leadframe, another semiconductor die, a PCB, etc.). Workpiece **120** is supported by support structure **104**. Docking station **112** is configured to receive a wire bonding tool replacement system **130**. Docking station **112** includes locator pins **112b** configured to guide a wire bonding tool replacement system **130** into the correct position during installation of wire bonding tool replacement system **130** on wire bonding system **100**.

[0033] FIG. **1B** illustrates wire bonding system **100**, wire bonding tool replacement system **130**, and a robot arm **154** that is carrying wire bonding tool replacement system **130**. Wire bonding tool replacement system **130** is configured to remove and replace wire bonding tool **106**. A full description of exemplary wire bonding tool replacement systems can be found in U.S. patent application Ser. No. 18/947,365 and U.S. patent application Ser. No. 18/947,209, the contents of which are incorporated herein by reference. A general description of the wire bonding tool replacement system can be found herein in connection with FIGS. **3A-3B**.

[0034] FIG. **1B** illustrates wire bonding tool replacement system **130** including a handle **130c** and a connector **130a**. Handle **130c** can be engaged by robot arm **154**. Robot arm **154** is part of a service robot **150** that is described in detail herein in connection with several figure series (e.g., FIGS. **4A-4E**). Robot arm **154** is configured to engage handle **130c** in order to install wire bonding tool replacement system **130** on and/or remove wire bonding tool replacement system **130** from wire bonding system **100**. Connector **130a** is capable of forming an electrical connection to transmit power and/or data (e.g., position information, execution instructions, information logs, etc.) between wire bonding tool replacement system **130** and wire bonding system **100** with a corresponding connector **112a** of docking station **112** (see FIGS. **2A-2B**). FIG. **1C** illustrates wire bonding tool replacement system **130** in an installed position on docking station **112**.

[0035] FIGS. **2A-2B** illustrate details related to the alignment and installation of wire bonding tool replacement system **130** on docking station **112**. FIG. **2A** is a top view of docking station **112**, and FIG. **2B** is a corresponding bottom view of exemplary wire bonding tool replacement system **130**. Docking station **112** includes locator pins **112b** (see e.g., locator pin **112b** in FIGS. **1A** and **1B**).

Locator pins **112b** are used to guide wire bonding tool replacement system **130** into the correct position during installation as they interface with corresponding locator holes **130b** on wire bonding tool replacement system **130**. Locator pins **112b** and locator holes **130b** ensure that connector **130a** of wire bonding tool replacement system **130** aligns with a connector **112a** of docking station **112**. As shown, wire bonding tool replacement system **130** has a male connector, and docking station **112** has a female connector, but the invention is not so limited. Further, different and/or additional alignment structures are contemplated (e.g., guide rails, alignment slots, navigational software, etc.).

[0036] Although wire bonding system **100** has been illustrated and described as including a docking station, the invention is not limited to such. It is contemplated that a wire bonding system may have any suitable receiving portion configured to receive a wire bonding tool replacement system (e.g., a designated flat area, a shelf, a hanger for holding the wire bonding tool replacement system, or any other structure capable of supporting the wire bonding tool replacement system).

[0037] FIGS. 3A-3B illustrate an exemplary wire bonding tool replacement system **130** illustrating different features (with connector **130a**, locator holes **130b**, and handle **130c** not illustrated in FIGS. 3A-3B for simplicity). Wire bonding tool replacement system **130** includes a used wire bonding tool receptacle **130d**, a maintenance arm **130e**, and a wire bonding tool tray holder **130f**. Used wire bonding tool receptacle **130d** is configured for holding used wire bonding tools **106'** after they are removed from a wire bonding system. Maintenance arm **130e** is capable of extending outward to perform the actions of removing and replacing a wire bonding tool on a wire bonding system (e.g., gripping an installed wire bonding tool, releasing the installed wire bonding tool from a transducer, removing the installed wire bonding tool from the transducer, gripping a new wire bonding tool, installing the new wire bonding tool, etc.). Wire bonding tool tray holder **130f** is configured to receive a wire bonding tool tray **140**. Wire bonding tool tray **140** is configured to include a plurality of wire bonding tools **106**. Wire bonding tool tray **140** may include a single type of wire bonding tool, or it may include various types of wire bonding tools, depending on the requirements of the application.

[0038] FIG. 3A illustrates wire bonding tool replacement system **130** in a “ready” state with maintenance arm **130e** extended. In FIG. 3A, in this example “ready” state, used wire bonding tool receptacle **130d** is empty, and wire bonding tool tray **140** contains the maximum number of wire bonding tools **106** (e.g., as shown, the maximum number of wire bonding tools is eight). It should be understood that for a wire bonding tool replacement system **130** to be “ready” it does not need to have an empty used wire bonding tool receptacle **130d**, and/or it does not need to have a full wire bonding tool tray **140**. For example, in order to be “ready”, a wire bonding tool replacement system **130** simply needs to include at least one wire bonding tool **106** in wire bonding tool tray **140** to be provided to a wire bonding system.

[0039] FIG. 3B illustrates wire bonding tool replacement system **130** in an “depleted” state with maintenance arm **130e** retracted. In FIG. 3B, in this “depleted” state, used wire bonding tool receptacle **130d** includes a plurality of used wire bonding tools **106'**, and wire bonding tool tray **140** is empty (e.g., a plurality of wire bonding tool slots **142** of wire bonding tool tray **140** are empty).

[0040] Wire bonding tool replacement system **130** may be installed on a single wire bonding system for a long period of time to automatically replace a wire bonding tool of the single wire bonding system when needed. However, wire bonding tool replacement system **130** is also capable of serving an array of wire bonding systems (e.g., an array including multiple wire bonding systems **100**) in a factory setting. The wire bonding systems in an exemplary wire bonding array may be configured to calculate an interval at which wire bonding tools need to be replaced (e.g., using a computer on the wire bonding system). Such a calculation may include factors such as a number of bonds, an amount of time, a change in process type/step, an expected tool life, a bond performance evaluation, environmental factors, etc. A wire bonding array also includes a service

robot configured to install, and remove, the wire bonding tool replacement system from each of the wire bonding systems. The service robot is capable of moving a wire bonding tool replacement system directly between wire bonding systems and/or between wire bonding systems and a central wire bonding tool replacement system storage area (e.g., a maintenance station, described herein). [0041] For example, when a wire bonding system requires a new wire bonding tool, it may (i) send a signal to a central computer of the wire bonding array, (ii) send a signal to a maintenance station, which would in turn instruct the service robot, and/or (iii) send a signal directly to the service robot. A central computer, maintenance station, or service robot may store identification and/or location information of each wire bonding system (along with the status of each wire bonding tool replacement system, that is “ready” or “depleted”). Such a central computer, maintenance station, or service robot may also be capable of sending and receiving instructions to and from components (e.g., wire bonding systems, a service robot, a maintenance station, etc.) of the wire bonding array. When a signal is received to change a wire bonding tool, the service robot may bring the wire bonding tool replacement system (that is known to be “ready”) to the wire bonding system that requires wire bonding tool replacement. After replacing the wire bonding tool, the wire bonding tool replacement system may then be moved to another wire bonding system that requires a wire bonding tool replacement or returned to a centralized location for storing and/or servicing wire bonding tool replacement systems (e.g., a maintenance station, described herein).

[0042] FIGS. 4A-4E, 5A-5E, 6A-6D, and 7A-7E illustrate exemplary processes for servicing a wire bonding array with a plurality of wire bonding tool replacement systems **130** by utilizing a service robot **150**. Specifically, FIGS. 4A-4E illustrate service robot **150** moving one of the plurality of wire bonding tool replacement systems **130** from a maintenance station **118** to one of the plurality of wire bonding systems **100**. FIGS. 5A-5E illustrate service robot **150** moving one of the plurality of wire bonding tool replacement systems **130** directly between ones of the plurality of wire bonding systems **100**. FIGS. 6A-6D illustrate service robot **150** moving one of the plurality of wire bonding tool replacement systems **130** from one of the plurality of wire bonding systems **100** to maintenance station **118**. FIGS. 7A-7E illustrate service robot **150** moving another of the plurality of wire bonding tool replacement systems **130** to another of the plurality of wire bonding systems **100**.

[0043] Referring to FIG. 4A, an exemplary wire bonding array **10** is shown. Wire bonding array **10** includes a plurality of wire bonding systems **100**, a maintenance station **118**, and a service robot **150**. Each of the plurality of wire bonding systems **100** includes a bond head assembly for bonding a wire to a workpiece and a support structure for supporting the workpiece (e.g., see FIGS. 1A-1C). Each of the plurality of wire bonding systems **100** further includes a docking station **112** (e.g., see FIGS. 2A-2B), each docking station **112** configured to receive any one of a plurality of wire bonding tool replacement systems **130**. Maintenance station **118** includes the plurality of wire bonding tool replacement systems **130** (e.g., see FIGS. 3A-3B). Maintenance station **118** further includes a plurality of wire bonding tool trays **140**. Maintenance station **118** also includes a robot charging station **180**. Service robot **150** includes a robot base **152** and robot arm **154**. Robot base **152** carries service robot **150** relative to the plurality of wire bonding systems **100**. Service robot **150** is configured to move throughout the wire bonding array **10**, e.g., by treads, wheels, overhead cables, a track, etc., via robot base **152**.

[0044] In FIG. 4B, service robot **150** has moved away from robot charging station **180**. Service robot **150** is in the process of securing wire bonding tool replacement system **130** using robot arm **154** by gripping a handle of wire bonding tool replacement system **130** (e.g., handle **130c** of FIGS. 1B and 1C). In FIG. 4C, service robot **150** has moved with wire bonding tool replacement system **130** to a position adjacent to one of the plurality of wire bonding systems **100**. Service robot **150** is aligning wire bonding tool replacement system **130** with docking station **112** of wire bonding system **100**. It is contemplated that service robot **150** may carry wire bonding tool replacement system **130** in any way during transit (e.g., by gripping handle **130c** during transit, by placing wire

bonding tool replacement system **130** on robot base **152** during transit, etc.). In FIG. 4C, the removal of wire bonding tool replacement system **130** has exposed a docking station **112** of maintenance station **118**. Maintenance station **118** includes a plurality of docking stations **112** that may correspond to the number of wire bonding tool replacement systems **130**. Each of the plurality of docking stations **112** of maintenance station **118** may be configured to receive each of the wire bonding tool replacement systems **130**. Each of the plurality of docking stations **112** of maintenance station **118** may include connections for power and/or data to facilitate communication between each of the plurality of wire bonding tool replacement systems **130** and a computer included in wire bonding array **10**.

[0045] In FIG. 4D, service robot **150** is installing wire bonding tool replacement system **130** onto docking station **112** of wire bonding system **100**. In FIG. 4E, wire bonding tool replacement system **130** has been installed on docking station **112** of wire bonding system **100**, and service robot **150** has returned to robot charging station **180**. Wire bonding tool replacement system **130** can begin the wire bonding tool replacement operation (i.e., the process of replacing the wire bonding tool installed on the wire bonding system with a different wire bonding tool **106**).

[0046] FIG. 5A illustrates wire bonding array **10** in the same state as FIG. 4E. However, in FIG. 5A, wire bonding tool replacement system **130** that was placed on wire bonding system **100** has completed the replacement of the wire bonding tool and is ready to be removed. Additionally, another of the plurality of wire bonding systems **100** requires replacement of its wire bonding tool. FIG. 5B illustrates service robot **150** securing wire bonding tool replacement system **130** with robot arm **154**. FIG. 5C illustrates service robot **150** aligning wire bonding tool replacement system **130** with the another of the plurality of wire bonding systems **100**. In FIG. 5D, service robot **150** is installing wire bonding tool replacement system **130** onto docking station **112** of the another of the plurality of wire bonding systems **100**. In FIG. 5E, wire bonding tool replacement system **130** has been installed on docking station **112** of the another of the plurality of wire bonding systems **100**, and service robot **150** has returned to robot charging station **180**. Wire bonding tool replacement system **130** can begin the wire bonding tool replacement.

[0047] FIG. 6A illustrates wire bonding array **10** in the same state as FIG. 5E. However, wire bonding tool replacement system **130** has completed the replacement of the wire bonding tool and is ready to be removed. FIG. 6B illustrates service robot **150** securing wire bonding tool replacement system **130** with robot arm **154**. In FIG. 6C, service robot **150** has removed wire bonding tool replacement system **130** from wire bonding system **100**. Service robot **150** is placing wire bonding tool replacement system **130** on to docking station **112** of maintenance station **118**. In FIG. 6D, wire bonding tool replacement system **130** has been returned to docking station **112** of maintenance station **118**, and service robot **150** has returned to robot charging station **180**. Wire bonding tool replacement system **130** can wait until another of the plurality of wire bonding systems **100** is ready for a wire bonding tool change, or it can be serviced by maintenance station **118** (e.g., see FIGS. 9A-9H, discussed in more detail herein).

[0048] FIG. 7A illustrates wire bonding array **10** in the same state as FIG. 4E. However, wire bonding tool replacement system **130** has not yet finished replacing the wire bonding tool of wire bonding system **100**. Further, another of the plurality of wire bonding systems **100** is ready to have its wire bonding tool replaced. In FIG. 7B, service robot **150** is securing another of the wire bonding tool replacement systems (i.e., wire bonding tool replacement system **130**). FIG. 7C illustrates service robot **150** aligning wire bonding tool replacement system **130** with wire bonding system **100**. In FIG. 7D, service robot **150** is installing wire bonding tool replacement system **130** onto docking station **112** of wire bonding system **100**. In FIG. 7E, wire bonding tool replacement system **130** has been installed on wire bonding system **100**, and service robot **150** has returned to robot charging station **180**.

[0049] Although FIGS. 4A-4E, 5A-5E, 6A-6D, and 7A-7E illustrate a wire bonding array with a trackless service robot (e.g., where service robot **150** uses wheels, treads, etc. to move around wire

bonding array **10**), the invention is not limited to such. It is contemplated that a service robot on a track (e.g., a track in a factory floor, an elevated track, etc.) may be beneficial for reducing the footprint of a wire bonding array. For example, FIGS. **8A-8C** illustrate various embodiments of the invention wherein the service robot travels along a track.

[0050] In particular, FIG. **8A** illustrates an overhead view of an exemplary wire bonding array **20**. Wire bonding array **20** includes a plurality of wire bonding systems **100**, a robot support structure **168**, and a maintenance station **118**. Robot support structure **168** supports a service robot **160** on a track **166**. Similar to service robot **150**, service robot **160** includes a robot base **162** and a robot arm **164**. Robot arm **164** is configured to engage handle **130c** (e.g., see FIGS. **1B** and **1C**) in order to install wire bonding tool replacement system **130** on and/or remove wire bonding tool replacement system **130** from wire bonding system **100**. Service robot **160** rides along track **166**. Service robot **160** is configured to move at least one wire bonding tool replacement system **130** between each of the plurality of wire bonding systems **100** and maintenance station **118**. Maintenance station **118** is described in additional detail herein.

[0051] FIG. **8B** illustrates a side view of wire bonding array **20**. In particular, FIG. **8B** illustrates that track **166** is elevated such that it supports robot base **162** above at least a portion of each of the plurality of wire bonding systems **100** (e.g., above docking station **112**, or any other receiving portion configured to receive a wire bonding tool replacement system). In FIG. **8B**, robot support structure **168** resembles an I-beam situated between wire bonding systems **100**, which are arranged back-to-back (although the invention is not limited to a back-to-back arrangement). By elevating service robot **160**, it is possible to arrange the plurality of wire bonding systems **100** such that there is a minimal amount of space between each of the plurality of wire bonding systems **100**. Maintenance station **118** may also be placed in closer proximity to the plurality of wire bonding systems **100**.

[0052] It is further contemplated that the space between each of the plurality of wire bonding systems **100** can be further reduced by suspending the robot support structure above the plurality of wire bonding systems **100**. FIG. **8C** illustrates such a wire bonding array **30**, including a plurality of wire bonding systems **100** and a robot support structure **178** including a track **176** that is suspended above the plurality of wire bonding systems **100** (e.g., suspended from a ceiling of a factory by cables, trusses, or other structure). Robot support structure **178** supports a service robot **170** that includes a robot base **172** and a robot arm **174** on track **176**. Track **176** supports robot base **172** above at least a portion of each of the plurality of wire bonding systems **100** (e.g., above docking station **112**, or any other receiving portion configured to receive a wire bonding tool replacement system). Wire bonding array **30** may also include a maintenance station (e.g., see maintenance station **118** in FIG. **8A**). By suspending service robot **170** above the remainder of wire bonding array **30**, each of the plurality of wire bonding systems **100** may be placed at a minimal distance from one another.

[0053] Although service robots **160** and **170** are illustrated and described as being supported by a track, the invention is not limited thereto. It is contemplated that a base of a service robot could be elevated and supported by any other structure, such as a cable or a plurality of cables, a gantry, a conveyor belt, a linear motor, a ball screw, etc.

[0054] As wire bonding tool replacement systems serve the wire bonding array, they use up their supply of new wire bonding tools and collect used wire bonding tools (e.g., see FIG. **3B**). Once a wire bonding tool replacement system runs out of new wire bonding tools, it may be reloaded. Such reloading may be done manually by human operators, or may be done automatically (e.g., at the maintenance station). For example, once an empty wire bonding tool replacement system is returned to the maintenance station (e.g., installed on a docking station of the maintenance station by a service robot, as in FIGS. **6A-6D**), a replacement assembly can (i) remove used wire bonding tools from the wire bonding tool replacement system, and/or (ii) provide new wire bonding tools to the wire bonding tool replacement system. FIGS. **9A-9H** illustrate such a process.

[0055] FIG. 9A illustrates maintenance station **118**. Maintenance station **118** includes a plurality of wire bonding tool replacement systems **130**, a plurality of wire bonding tool trays **140**, a replacement assembly **118a**, and a waste receptacle **118b**. Each of the plurality of wire bonding tool replacement systems **130** is installed on one of a plurality of docking stations **112** (not illustrated). One of the wire bonding tool replacement systems **130** is illustrated in the “depleted” state (e.g., see FIG. 3B, described herein). Each of the plurality of wire bonding tool replacement systems **130** is configured for installation on, and removal from, a wire bonding system (e.g., see FIGS. 1B, 1C, 2A, 2B, and 4A-4E). Each of the plurality of wire bonding tool trays **140** includes a plurality of wire bonding tools **106**.

[0056] Replacement assembly **118a** is configured to remove waste from each of the plurality of wire bonding tool replacement systems **130**. Such waste may include (i) used wire bonding tools **106'** that are stored in used wire bonding tool receptacle **130d**, and/or (ii) an empty wire bonding tool tray **140**. Replacement assembly **118a** may include a gripper, a clamp, a suction device, or other structure known in the art for selectively moving items.

[0057] In FIG. 9B, replacement assembly **118a** has moved to wire bonding tool replacement system **130** to remove an empty wire bonding tool tray **140** from wire bonding tool replacement system **130**. In FIG. 9C, replacement assembly **118a** has taken empty wire bonding tool tray **140** out of wire bonding tool replacement system **130** and moved it to waste receptacle **118b**. In FIG. 9D, wire bonding tool tray **140** has been deposited in waste receptacle **118b**, and replacement assembly **118a** has returned to wire bonding tool replacement system **130** to remove the used wire bonding tools **106'**. In FIG. 9E, replacement assembly **118a** has removed used wire bonding tools **106'** (e.g., with or without used wire bonding tool receptacle **130d**) from wire bonding tool replacement system **130** and brought them to waste receptacle **118b**. In FIG. 9F, used wire bonding tools **106'** have been deposited in waste receptacle **118b**.

[0058] Additionally, FIG. 9F illustrates replacement assembly **118a** in the process of picking up one of the plurality of wire bonding tool trays **140** to load into wire bonding tool replacement system **130**. In FIG. 9G, replacement assembly **118a** is shown loading wire bonding tool tray **140** into wire bonding tool replacement system **130**. FIG. 9H illustrates replacement assembly **118a** in a home position, and wire bonding tool replacement system **130** in a “ready” state.

[0059] As will be appreciated by those skilled in the art, replacement assembly **118a** may include multiple different mechanisms for carrying out the purpose of removing waste and/or loading a wire bonding tool tray (e.g., one or more mechanisms for carrying out the following: for removing waste from used wire bonding tool receptacle **130d**; for removing empty wire bonding tool tray **140**; and/or for loading a wire bonding tool tray **140**).

[0060] Although maintenance station **118** has been illustrated and described as using a replacement assembly for certain functions (e.g., discarding waste, servicing and/or refreshing wire bonding tool replacement systems **130**, etc.), the invention is not so limited. For example, operators may perform certain of these functions. Thus, a replacement assembly may be different from that shown herein, and in fact may be omitted in certain applications.

[0061] Although maintenance station **118** has been illustrated and described as including only wire bonding tool replacement systems and wire bonding tool trays, the invention is not limited to such. It is contemplated that a maintenance station could include additional elements used in wire bonding operations. For example, a maintenance station could include workpieces (e.g., workpieces to be bonded, completed workpieces, magazines for holding a plurality of workpieces to be bonded or completed workpieces, etc.), wire spools, or other consumables, each of which able to be distributed by a service robot to the plurality of wire bonding systems (e.g., wherein a singular service robot is configured to move and/or install each of the consumables stored at the maintenance station).

[0062] FIGS. 10-11 are flow diagrams describing methods related to the above-described wire bonding arrays. As is understood by those skilled in the art, certain steps included in the flow

diagram may be omitted; certain additional steps may be added; and the order of steps may be altered from the order illustrated—all within the scope of the invention.

[0063] Referring now to FIG. **10**, at Step **1000**, a wire bonding array is provided (e.g., wire bonding array **10** of FIGS. **4A-4E**, **5A-5E**, **6A-6D**, and **7A-7E**, wire bonding array **20** of FIGS. **8A** and **8B**, wire bonding array **30** of FIG. **8C**, etc.). The wire bonding array includes a plurality of wire bonding systems, each of the plurality of wire bonding systems includes a bond head assembly for bonding a wire to a workpiece and a support structure for supporting the work piece, the bond head assembly including a wire bonding tool (e.g., see wire bonding system **100** in FIGS. **1A-1C**). The wire bonding array also includes a wire bonding tool replacement system configured for installation on, and removal from, each of the plurality of wire bonding systems (e.g., see wire bonding tool replacement system **130** of FIGS. **3A-3B**). The wire bonding array further includes a service robot configured to install the wire bonding tool replacement system on, and remove the wire bonding tool replacement system from, each of the plurality of wire bonding systems (e.g., see service robot **150** of FIGS. **4A-4E**, **5A-5E**, **6A-6D**, and **7A-7E**).

[0064] At Step **1002**, the wire bonding tool replacement system is carried by the service robot to one of the plurality of wire bonding systems (e.g., see FIG. **4C**). At Step **1004**, the wire bonding tool replacement system is installed on the one of the plurality of wire bonding systems using the service robot (e.g., see FIG. **4D**). At optional Step **1006**, the wire bonding tool replacement system is removed from the one of the plurality of wire bonding systems with the service robot after Step **1004** (e.g., see FIGS. **5B** and **6B**).

[0065] After the wire bonding tool replacement system is removed from the wire bonding system in optional Step **1006**, one of two things may occur, either optional Steps **1008** and **1010** or optional Steps **1012** and **1014**. In optional Step **1008**, the wire bonding tool replacement system is carried to another of the plurality of wire bonding systems with the service robot (e.g., see FIG. **5C**) after optional Step **1006**. In optional Step **1010**, the wire bonding tool replacement system is installed on the another of the plurality of wire bonding systems after optional Step **1008** (e.g., see FIG. **5D**). In optional Step **1012**, the wire bonding tool replacement system is carried to a maintenance station by the service robot after optional Step **1006** (e.g., see FIG. **6C**). In optional Step **1014**, the wire bonding tool replacement system is installed on a docking station of the maintenance station after Step **1012** (e.g., see FIGS. **6C** and **6D**).

[0066] Referring now to FIG. **11**, at Step **1100** a maintenance station is provided e.g., see FIGS. **9A-9H**). The maintenance station includes a plurality of wire bonding tool replacement systems, each of the plurality of wire bonding tool replacement systems configured for installation on, and removal from, a wire bonding system. The maintenance station also includes a plurality of wire bonding tool trays, each of the plurality of wire bonding tool trays including a plurality of wire bonding tools (e.g., plurality of wire bonding tool trays **140** of FIGS. **9A-9H**). The maintenance station further includes a replacement assembly configured to load one of the plurality of wire bonding tool trays into one of the plurality of wire bonding tool replacement systems (e.g., replacement assembly **118a** of FIGS. **9A-9H**). At Step **1102**, one of the plurality of wire bonding tool trays is installed in one of the plurality of wire bonding tool replacement systems using the replacement assembly (see, e.g., FIGS. **9F-9G**). At optional Step **1104**, an empty wire bonding tool tray is removed from the one of the plurality of wire bonding tool replacement systems with the replacement assembly prior to step **1102** (e.g., see FIGS. **9B-9C**). In optional step **1106**, at least one used wire bonding tool is removed from a used wire bonding tool receptacle of the one of the plurality of wire bonding tool replacement systems with the replacement assembly (e.g., see FIGS. **9D-9E**).

[0067] Although the invention is illustrated and described herein as including only one type of wire bonding tool replacement system, the invention is not intended to be limited thereto. As will be appreciated by those skilled in the art, wire bonding tool replacement systems may be configured in different ways, including compatibility with different wire bonding tools, or including additional

features, and each wire bonding tool replacement system included in the wire bonding array need not be the same.

[0068] Although the invention is illustrated and described herein as including only a single service robot, the invention is not intended to be limited thereto. A single service robot was illustrated and described solely for clarity and conciseness. It is contemplated that a wire bonding array would likely include a plurality of service robots.

[0069] Although the invention is illustrated and described herein with reference to specific embodiments, the invention is not intended to be limited to the details shown. Rather, various modifications may be made in the details within the scope and range of equivalents of the claims and without departing from the invention.

Claims

1. A wire bonding array comprising: a plurality of wire bonding systems, each of the plurality of wire bonding systems including (i) a bond head assembly for bonding a wire to a workpiece and (ii) a support structure for supporting the workpiece; a wire bonding tool replacement system configured for installation on, and removal from, each of the plurality of wire bonding systems; and a service robot configured to install the wire bonding tool replacement system on, and remove the wire bonding tool replacement system from, each of the plurality of wire bonding systems.
2. The wire bonding array of claim 1 wherein each of the plurality of wire bonding systems includes a docking station configured to receive the wire bonding tool replacement system.
3. The wire bonding array of claim 2 wherein the docking station includes a connector for providing power and data to the wire bonding tool replacement system.
4. The wire bonding array of claim 2 wherein the docking station includes locator pins for aligning the wire bonding tool replacement system during installation.
5. The wire bonding array of claim 1 further comprising at least one additional wire bonding tool replacement system configured for installation on, and removal from, each of the plurality of wire bonding systems.
6. The wire bonding array of claim 1 wherein the service robot includes a base for moving the service robot with respect to the plurality of wire bonding systems, the base being positioned above a receiving portion of each of the plurality of wire bonding systems configured to receive the wire bonding tool replacement system.
7. A wire bonding array comprising: a plurality of wire bonding systems, each of the plurality of wire bonding systems including (i) a bond head assembly for bonding a wire to a workpiece and (ii) a support structure for supporting the workpiece; a maintenance station including a plurality of wire bonding tool replacement systems, each of the plurality of wire bonding tool replacement systems being configured for installation on, and removal from, each of the plurality of wire bonding systems; and a service robot configured to move ones of the plurality of wire bonding tool replacement systems between the maintenance station and ones of the plurality of wire bonding systems.
8. The wire bonding array of claim 7 wherein each of the plurality of wire bonding systems includes a docking station configured to receive each of the plurality of wire bonding tool replacement systems.
9. The wire bonding array of claim 7 wherein the maintenance station includes a plurality of docking stations, each of the plurality of docking stations configured to receive each of the plurality of wire bonding tool replacement systems.
10. The wire bonding array of claim 7 wherein the maintenance station includes a replacement assembly configured to remove a waste from each of the plurality of wire bonding tool replacement systems, the waste including at least one of (i) an empty wire bonding tool tray configured for holding a plurality of wire bonding tools and (ii) at least one used wire bonding tool.

- 11.** The wire bonding array of claim 10 wherein the maintenance station further includes a receptacle for receiving the waste from the plurality of wire bonding tool replacement systems from the replacement assembly.
 - 12.** The wire bonding array of claim 7 wherein the maintenance station further includes a plurality of wire bonding tool trays, each of the plurality of wire bonding tool trays including a plurality of wire bonding tools.
 - 13.** The wire bonding array of claim 12 wherein the maintenance station includes a replacement assembly configured to load one of the plurality of wire bonding tool trays into one of the plurality of wire bonding tool replacement systems.
 - 14.** The wire bonding array of claim 7 wherein the maintenance station further includes a charging station for the service robot.
 - 15.** The wire bonding array of claim 7 wherein the service robot is further configured to move ones of the plurality of wire bonding tool replacement systems directly between ones of the plurality of wire bonding systems.
 - 16.** The wire bonding array of claim 7 wherein the service robot includes a base for moving the service robot with respect to the plurality of wire bonding systems, the base being positioned above a receiving portion of each of the plurality of wire bonding systems configured to receive each of the plurality of wire bonding tool replacement systems.
 - 17.** A maintenance station for servicing a wire bonding array, the maintenance station comprising: a plurality of wire bonding tool replacement systems, each of the plurality of wire bonding tool replacement systems being configured for installation on, and removal from, a wire bonding system.
 - 18.** The maintenance station of claim 17 further comprising a plurality of docking stations configured to receive any one of the plurality of wire bonding tool replacement systems.
 - 19.** The maintenance station of claim 17 further comprising a replacement assembly configured to remove a waste from each of the plurality of wire bonding tool replacement systems, the waste including at least one of (i) an empty wire bonding tool tray configured for holding a plurality of wire bonding tools and (ii) at least one used wire bonding tool.
 - 20.** The maintenance station of claim 19 further comprising a receptacle for receiving the waste from the plurality of wire bonding tool replacement systems.
 - 21-38** (canceled)
-